Day 167: The Report Card: SHAP and LIME from Scratch, and the Week's Six Tools on One Model

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 7 | Complete study guide

Watch: IML - 04 Shapley - 03 SHAP (SHapley Additive exPlanation) Values (Statistical Learning and Data Science) | "Why Should I Trust You?" Explaining the Predictions of Any Classifier (KDD2016 video)

Script: `day167_report_card.py` · every number in this guide is copied by program from `day167_run_output.txt`, the real output of that script.

Days 161–166 each built one way of checking a model. Today they are used together, once, on a model chosen before Adult's test file was opened. Boosting reached accuracy 0.8691 [0.8636, 0.8744] on the 15060 test rows and beat logistic regression clearly (McNemar exact p 4.52e-39). Its sex gaps from Day 166 held up on the test file. SHAP and LIME are built from scratch: exact Shapley values match a closed form to 16 decimal places, KernelSHAP's sampling error is small (0.0035 at 1,000 coalitions), and the first draft's real weakness was the 50-row background, which moved values by up to 0.4536.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

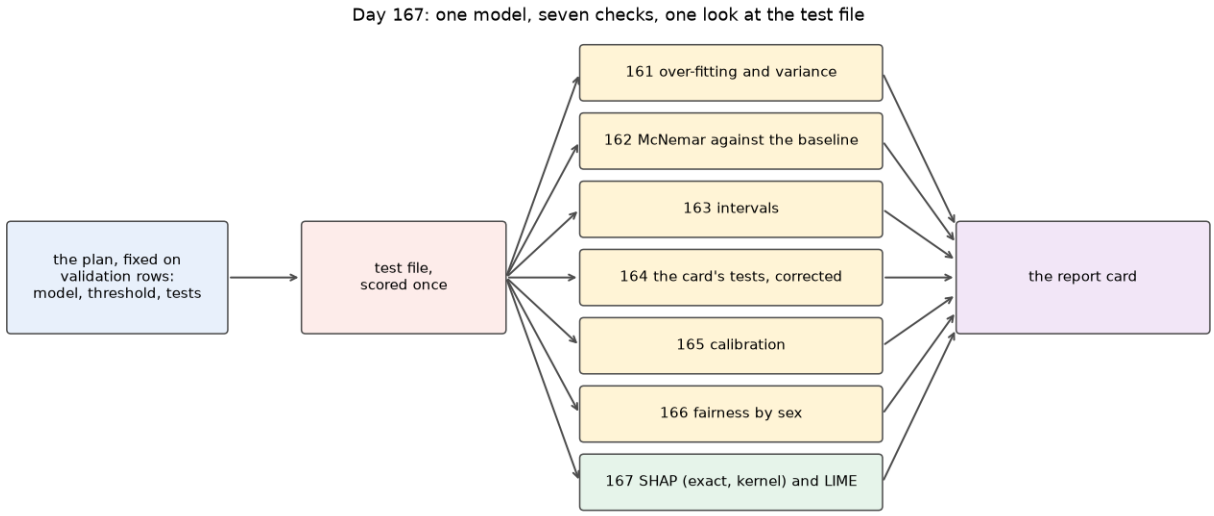
A **report card** checks one model in several independent ways and writes the results down together. It is only honest if the model, the threshold and the tests are fixed **before** the test data is looked at (**pre-registration**); otherwise the test file quietly becomes another validation set.

SHAP explains one prediction by sharing out the difference between it and an average prediction among the features, using the **Shapley value** from game theory: each feature gets its average contribution over every order in which features could be added. **LIME** explains one prediction by fitting a small weighted linear model to the model's answers on made-up rows near it.

REAL-WORLD SCENARIO

Before a credit model goes live, a review board asks: how accurate is it, and how sure are you? Is it better than what we use now? Is it over-fitted? Are its probabilities right? Does it treat groups differently? Which inputs drive a given decision? Each answer comes from one of this week's days.

SYSTEM DIAGRAM



The plan is fixed on the validation rows; the test file is scored once; seven checks feed one card.

The data and split are Week 24's: UCI Adult's training file split into 24,130 fitting rows and 6,032 validation rows (Day 161), and Adult's separate test file, untouched until today. Both models are trained on the fitting rows only. Today has no exercise file: it imports the six built this week (`day161_exercise.py` to `day166_exercise.py`) and uses them unchanged.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The plan, then one look at the test file

DEFINITION — start here, no prior knowledge assumed

Section 0 prints the evidence from the validation rows and the plan: boosting is the model, logistic regression the baseline, the threshold is 0.5, and six named tests will be corrected with Holm's method. Only then are the test rows scored.

Performance gets intervals (Day 163): Wilson's for accuracy, a 1,000-resample bootstrap for the AUC. The comparison with the baseline uses McNemar's test (Day 162), which looks only at the rows where the two models disagree.

REAL-WORLD SCENARIO

A drug trial registers its primary outcome before unblinding. If it reported whichever of twenty outcomes looked best afterwards, its p-value would mean little.

ACTUAL OUTPUT

```
  validation, logistic: accuracy 0.8408, AUC 0.8972, ECE 0.0149
  validation, boosting: accuracy 0.8662, AUC 0.9258, ECE 0.0096
  model: boosting (more accurate on validation, and calibrated); baseline: logistic regression
  threshold 0.5; the six tests in section 6 are corrected with Holm at 0.05 (Benjamini-Hochberg shown beside it)
  the test file is scored once, below; nothing is changed after seeing it
Actual output from this lesson's run.
```

ACTUAL OUTPUT

```
  15060 test rows, 0.2457 earn above $50K
  fast_auc against Day 117's roc_auc on these rows: 0.9240055077 vs 0.9240055077
  boosting: accuracy 0.8691 [0.8636, 0.8744]; AUC 0.9240 [0.9197, 0.9285]
  logistic: accuracy 0.8421 [0.8362, 0.8478]; AUC 0.8970 [0.8917, 0.9023]
Actual output from this lesson's run.
```

ACTUAL OUTPUT

```
  only boosting wrong 291, only logistic wrong 697: chi-square 166.02, p 5.48e-38, exact p 4.52e-39
  AUC difference +0.0270 [+0.0244, +0.0296] (paired bootstrap)
Actual output from this lesson's run.
```

What it means

On validation boosting had accuracy 0.8662 against 0.8408, so it was chosen. On the test file it reached 0.8691 [0.8636, 0.8744] and AUC 0.9240 [0.9197, 0.9285]; logistic regression 0.8421 and 0.8970. The test file agreed with validation, as it should when the plan is fixed first.

Of the rows where the two models disagreed, boosting alone was wrong on 291 and logistic regression alone on 697: exact p 4.52e-39. The paired AUC difference was +0.0270 [+0.0244, +0.0296].

One practical change: Day 117's AUC averages tied ranks with a loop over every distinct score, about half a second on 15,060 distinct scores, so 1,000 resamples would take minutes. `fast_auc` does the same with one sort; on the test predictions both gave 0.9240055077.

WHY CHOOSE THIS?

A fixed plan makes the test result an honest estimate; intervals and a paired test show how sure it is.

WHY NOT THIS? (TRADEOFFS)

The test file can be used once; anything changed after looking at it needs new data to check.

2. Over-fitting and variance (Day 161)

DEFINITION — start here, no prior knowledge assumed

Two checks. First, each final model's Brier score on its own training rows against the test file: a large gap means it memorised. Second, each model is retrained on four disjoint quarters of the fitting rows and Day 161's decomposition splits their test error into bias² + noise and variance.

REAL-WORLD SCENARIO

A student who scores 95% on homework they have seen and 70% on a new exam has learned the answers, not the subject.

ACTUAL OUTPUT

```
  logistic: Brier on its training rows 0.1084, on the test file 0.1089
  boosting: Brier on its training rows 0.0870, on the test file 0.0917
  logistic trained on quarters (6033 rows each): Brier 0.1095 = bias^2 + noise 0.1087 + variance 0.0006
  boosting trained on quarters (6033 rows each): Brier 0.0954 = bias^2 + noise 0.0921 + variance 0.0025
  (quarter models took 15 s)
Actual output from this lesson's run.
```

What it means

Logistic regression barely over-fits (0.1084 on its training rows, 0.1089 on test). Boosting shows a small gap (0.0870 against 0.0917), still far better than logistic regression on test.

On quarter-sized training sets, boosting's variance was 0.0025 against logistic regression's 0.0006: the flexible model depends more on which rows it saw. Its lower bias more than paid for that (Brier 0.0954 against 0.1095). These are quarter-sized models; the final model, trained on four times as many rows, was not retrained here, so its own variance is not measured.

WHY CHOOSE THIS?

Training-against-test and the decomposition say whether more data or a simpler model would help.

WHY NOT THIS? (TRADEOFFS)

Variance needs several retrained models, which costs time and uses smaller training sets than the final model.

3. Calibration and fairness on the test file (Days 165 and 166)

DEFINITION — start here, no prior knowledge assumed

Calibration: the ECE on the test rows, against the ECE a perfectly calibrated model would show by chance (200 simulated worlds), overall and per sex.

Fairness: Day 166's four rates per sex at 0.5, the gaps with bootstrap intervals, and the validation gaps beside them, to see whether they held.

REAL-WORLD SCENARIO

A weather service checks its rain forecasts each season, and also checks them separately for the coast and the mountains: a forecast can be right on average and wrong in one region.

ACTUAL OUTPUT

```
  boosting: ECE 0.0118 (chance floor 0.0060); mean probability 0.2467 against a rate of 0.2457
  logistic: ECE 0.0093 (chance floor 0.0065); mean probability 0.2482 against a rate of 0.2457
  boosting, Female: ECE 0.0089 (chance floor 0.0076); mean probability 0.1157 against 0.1134
  boosting, Male  : ECE 0.0163 (chance floor 0.0082); mean probability 0.3101 against 0.3097
Actual output from this lesson's run.
```

ACTUAL OUTPUT

```
  Female: 4913 test rows, base rate 0.1134; selected 0.0804, TPR 0.5727, FPR 0.0174, precision 0.8076
  Male  : 10147 test rows, base rate 0.3097; selected 0.2680, TPR 0.6688, FPR 0.0881, precision 0.7731
  gap men - women in sel: test +0.1876 [+0.1759, +0.1997]; validation +0.1769
  gap men - women in tpr: test +0.0961 [+0.0536, +0.1414]; validation +0.0713
  gap men - women in fpr: test +0.0706 [+0.0631, +0.0783]; validation +0.0670
  gap men - women in ppv: test -0.0345 [-0.0756, +0.0053]; validation -0.0416
  women's selection rate / men's = 0.300
Actual output from this lesson's run.
```

What it means

Boosting's average probability matched the rate almost exactly (0.2467 against 0.2457), but its ECE, 0.0118, was about twice its chance floor (0.0060); for men 0.0163 against 0.0082. On validation it had sat at its floor. The error is small, and the plan does not allow fitting a calibrator now: Day 165 showed it must be fitted on rows the model never saw, and the test file is spent.

The fairness gaps held: selection +0.1876 on test against +0.1769 on validation, TPR +0.0961 [+0.0536, +0.1414] against +0.0713. Precision again did not differ detectably (-0.0345 [-0.0756, +0.0053]). Women's selection rate was 0.300 of men's.

WHY CHOOSE THIS?

Checks on the final model what Days 165 and 166 found on validation, with nothing tuned on the result.

WHY NOT THIS? (TRADEOFFS)

A problem found here cannot be fixed and re-checked on the same rows.

4. The card's six tests, corrected (Day 164)

DEFINITION — start here, no prior knowledge assumed

The card makes six yes/no claims: boosting beats logistic regression; test accuracy differs from validation accuracy (a check for drift); and the four sex gaps. Six tests at 0.05 would give a false alarm more often than 5% of the time, so Holm's method, chosen in the plan, decides; Benjamini-Hochberg and Bonferroni are shown beside it.

REAL-WORLD SCENARIO

A building inspector checks six things. If each check has a 5% chance of a false alarm, the report as a whole has about a 26% chance of at least one, if the checks are independent.

ACTUAL OUTPUT

```
  boosting vs logistic (McNemar)  : p 4.52e-39; no correction reject, Bonferroni reject, Holm reject, Benjamini-Hochberg
      reject
  accuracy, test vs validation    : p 5.81e-01; no correction keep, Bonferroni keep, Holm keep, Benjamini-Hochberg keep
  sel gap by sex                  : p 2.00e-156; no correction reject, Bonferroni reject, Holm reject, Benjamini-Hochberg
      reject
  tpr gap by sex                  : p 1.11e-05; no correction reject, Bonferroni reject, Holm reject, Benjamini-Hochberg
      reject
  fpr gap by sex                  : p 8.03e-53; no correction reject, Bonferroni reject, Holm reject, Benjamini-Hochberg
      reject
  ppv gap by sex                  : p 1.23e-01; no correction keep, Bonferroni keep, Holm keep, Benjamini-Hochberg keep
Actual output from this lesson's run.
```

What it means

Four of the six were significant under every rule. The drift test was not (p 5.81e-01): no evidence that accuracy on the test file differs from the validation rows. The precision gap was not either (p 1.23e-01). With p-values this far apart, the correction changed no decision here; it would matter for claims near 0.05.

WHY CHOOSE THIS?

The card's error rate is controlled as a whole, not claim by claim.

WHY NOT THIS? (TRADEOFFS)

Corrections cost power; listing the tests in advance matters as much as the correction itself.

5. SHAP from scratch

DEFINITION — start here, no prior knowledge assumed

A **coalition** is a set of features taken from the row being explained; every other feature takes its value from a **background** row, and the model's output is averaged over the background. That average is v(S). A feature's Shapley value is its added v when it joins a coalition, averaged over all coalitions with the Shapley weights. With 12 features there are 4,096 coalitions, few enough to compute exactly.

KernelSHAP samples coalitions instead and fits a weighted linear regression whose coefficients are the Shapley values; it is needed when there are too many features to list every coalition. Both explain log-odds, so the values add up.

REAL-WORLD SCENARIO

Three people share a taxi to different stops. The Shapley fare split charges each person their average extra cost over every order in which they could have joined the ride.

MATHEMATICS

$$\phi_j = \Sigma_{S \subseteq N \setminus \{j\}} |S|!\,(n - |S| - 1)!\,/\,n! \cdot [\, v(S \cup \{j\}) - v(S)\,]$$

$$v(S) = \text{mean over background rows } b \text{ of } f(x_S, b_{\text{not } S}) \qquad \Sigma_j \phi_j = f(x) - v(\varnothing)$$

$$\text{linear model: } \phi_j = \Sigma_{\text{columns } c \text{ of } j} w_c\,(x_c - \text{mean of } b_c)$$

IMPLEMENTATION

```
def shap_exact(f, x, bg, colfeat):
    v = coalition_values(f, x, bg, ALL, colfeat)
    code = ALL @ (1 << np.arange(NF)); size = ALL.sum(1)
    wt = np.array([factorial(s) * factorial(NF - s - 1) / factorial(NF) for s in range(NF)])
    phi = np.zeros(NF)
    for j in range(NF):
        S = code[~ALL[:, j]]
        phi[j] = np.sum(wt[size[S]] * (v[S + (1 << j)] - v[S]))
    return phi, v[0], v[-1]
```

ACTUAL OUTPUT

```
explaining log-odds; a feature 'absent' takes its value from one of 200 background rows (averaged over all 200)
logistic, test row 9516: largest difference from the closed form 1.1e-16; sum of values +1.2848 = f(x) - background mean
    +1.2848
logistic, test row 4987: largest difference from the closed form 2.2e-16; sum of values +3.1336 = f(x) - background mean
    +3.1336
logistic, test row 1106: largest difference from the closed form 2.2e-16; sum of values -1.5081 = f(x) - background mean
    -1.5081
boosting, test row 9516: sum of values +0.7106 = f(x) - background mean +0.7106; largest: occupation -0.740
boosting, test row 4987: sum of values +3.5672 = f(x) - background mean +3.5672; largest: age +0.878
boosting, test row 1106: sum of values -3.9277 = f(x) - background mean -3.9277; largest: age -1.365
boosting, test row 8157: sum of values -2.5127 = f(x) - background mean -2.5127; largest: age -0.974
boosting, test row 327: sum of values +0.8836 = f(x) - background mean +0.8836; largest: education-num +0.660
(exact values for 5 rows: 12 s, 4096 coalitions x 200 background rows each)
```

ACTUAL OUTPUT

```
  KernelSHAP with   50 coalitions: largest error against exact 0.0275, mean 0.0109 (5 rows x 2 seeds)
  KernelSHAP with  200 coalitions: largest error against exact 0.0073, mean 0.0048 (5 rows x 2 seeds)
  KernelSHAP with 1000 coalitions: largest error against exact 0.0035, mean 0.0021 (5 rows x 2 seeds)
  two backgrounds of  50 rows: exact values differ by up to 0.4536 (mean of the per-row largest 0.3855)
  two backgrounds of 200 rows: exact values differ by up to 0.0468 (mean of the per-row largest 0.0424)
  first draft, 50 background rows (seed 5): top three relationship 0.589, age 0.566, marital-status 0.461; capital-gain
      0.365, rank 5; the background has 3 rows with a capital gain, largest 99999
  first draft, 50 background rows (seed 7): top three capital-gain 0.634, relationship 0.629, age 0.566; capital-gain 0.634,
      rank 1; the background has 7 rows with a capital gain, largest 20051
  mean |SHAP value| over 200 test rows (151 s): relationship 0.599, age 0.582, marital-status 0.460, capital-gain 0.390,
      education-num 0.380, occupation 0.342, hours-per-week 0.338, sex 0.141, workclass 0.103, native-country 0.054,
      capital-loss 0.030, race 0.030
  check with a second background of 200 rows: relationship 0.617, age 0.554, marital-status 0.472, capital-gain 0.404,
      education-num 0.388, occupation 0.350, hours-per-week 0.340, sex 0.142, workclass 0.099, native-country 0.072,
      capital-loss 0.066, race 0.034; largest change 0.036
  mean SHAP value of sex           : women -0.194 (68 rows), men +0.114 (132 rows)
  mean SHAP value of relationship  : women -0.236 (68 rows), men +0.105 (132 rows)
  mean SHAP value of marital-status: women -0.229 (68 rows), men +0.142 (132 rows)
Actual output from this lesson's run.
```

What it means

The known answer. For logistic regression the exact values matched the closed form to 2.2e-16, rounding error, and every row's values added up to its prediction minus the background mean. This checks the coalition code against something derived independently.

KernelSHAP. Against the exact values on boosting, its largest error fell from 0.0275 with 50 coalitions to 0.0035 with 1,000.

The first draft's real problem was the background. It used 50 background rows. Two different sets of 50 changed the exact values by up to 0.4536, about a hundred times KernelSHAP's error, and changed the global ranking: capital-gain was 5th (0.365) with one background and 1st (0.634) with the other. The second background held 7 rows with a capital gain against 3 rows with a capital gain in the first; a likely reason, not tested here, is that when few people have a capital gain, how many happen to be in the background decides how much 'removing' one is worth. With 200 rows the difference fell to 0.0468, and a second 200-row background moved the global importances by at most 0.036.

What the model uses. Relationship (0.599), age (0.582) and marital status (0.460) contribute most on average. The sex column contributes -0.194 for women and +0.114 for men on average, and relationship and marital status, its proxies from Day 166, add more in the same direction. These describe the model's arithmetic, not the causal effect of anyone's sex.

WHY CHOOSE THIS?

SHAP's values add up exactly and are unique given the background; exact values are checkable when features are few.

WHY NOT THIS? (TRADEOFFS)

They depend on the background (choose it large and report it), cost many model calls, and say nothing about causes.

6. LIME from scratch

DEFINITION — start here, no prior knowledge assumed

LIME makes 1,000 random variants of the row: each feature is kept or replaced by a background row's value, at random. Variants closer to the row (fewer features changed) get more weight, through a kernel of a chosen width. A weighted linear regression of the model's output on the kept/changed indicators gives one coefficient per feature.

This version is simplified: it switches whole features on and off like SHAP, while the lime package by default bins numbers into quartiles. The settings tested are the random seed, the number of samples and the width.

REAL-WORLD SCENARIO

To understand how a thermostat responds, you nudge the dial a little in random directions and note what happens, trusting the nudges nearest the current setting most.

MATHEMATICS

$$\beta = \operatorname{argmin} \Sigma_z \, \pi(z) \, (f(z) - \beta_0 - \beta \cdot z)^2 + \lambda \, \Sigma_j \, \beta_j^2, \quad \pi(z) = \exp(-(\text{features changed}) / \text{width}^2)$$

IMPLEMENTATION

```
def lime(f, x, bg, colfeat, N=LIME_N, width=None, seed=0):
    rng = np.random.RandomState(seed); width = width or 0.75 * np.sqrt(NF)
    Z = rng.rand(N, NF) < 0.5; Z[0] = True
    H = np.where(Z[:, colfeat], x[None, :], bg[rng.randint(0, len(bg), N)])
    wt = np.exp(-(~Z).sum(1) / width ** 2)
    A = np.c_[np.ones(N), Z.astype(float)]
    R = 1e-3 * np.eye(NF + 1); R[0, 0] = 0.0
    return np.linalg.solve(A.T @ (A * wt[:, None]) + R, A.T @ (wt * f(H)))[1:]
```

ACTUAL OUTPUT

```
  test row 9516, exact SHAP: occupation -0.740, relationship +0.639, marital-status +0.609, age +0.325
  LIME, 1000 samples, 5 seeds: occupation -0.766 to -0.701; relationship +0.574 to +0.737; marital-status +0.524 to +0.709;
      age -0.046 to +0.466; largest spread across seeds 0.513 (age)
  LIME, 5000 samples, 5 seeds: occupation -0.813 to -0.729; relationship +0.588 to +0.713; marital-status +0.550 to +0.729;
      age +0.275 to +0.379; largest spread across seeds 0.179 (marital-status)
  width 0.50: occupation -0.663, relationship +0.545, marital-status +0.108, age +0.186; largest difference from SHAP 0.501
  width 1.00: occupation -0.773, relationship +0.573, marital-status +0.641, age +0.227; largest difference from SHAP 0.107
  width 2.60: occupation -0.730, relationship +0.605, marital-status +0.616, age +0.275; largest difference from SHAP 0.071
  width 10.00: occupation -0.721, relationship +0.614, marital-status +0.616, age +0.280; largest difference from SHAP 0.079
  on logistic regression (linear in these on/off features), LIME vs the closed-form SHAP: largest difference 0.073
  over 20 rows at the defaults: top-3 features shared with SHAP 2.55 of 3 on average (all 3 in 11 rows); correlation of the
      12 values 0.925 on average, lowest 0.811
Actual output from this lesson's run.
```

What it means

With 1,000 samples, five seeds gave age anywhere from -0.046 to +0.466 for the same row; exact SHAP said +0.325. Five times the samples narrowed the largest spread to 0.179. A narrow width (0.5) changed the explanation: marital status fell to +0.108 against SHAP's +0.609. At the usual width and wider, LIME stayed within about 0.08 of SHAP.

Over 20 rows at the defaults, LIME and KernelSHAP shared 2.55 of their top three features on average (all three in 11 rows), with a correlation of 0.925 on average and 0.811 at worst. LIME's answer is a function of its settings; run it with several seeds and report the spread.

WHY CHOOSE THIS?

LIME needs no weighting theory, works with any model, and its linear surrogate is easy to read.

WHY NOT THIS? (TRADEOFFS)

Its answer moves with the seed, the sample size and the width, and nothing makes its values add up.

7. The report card, and the week

DEFINITION — start here, no prior knowledge assumed

Section 9 writes the results in one place. Every line of it comes from a check planned before the test file was opened.

REAL-WORLD SCENARIO

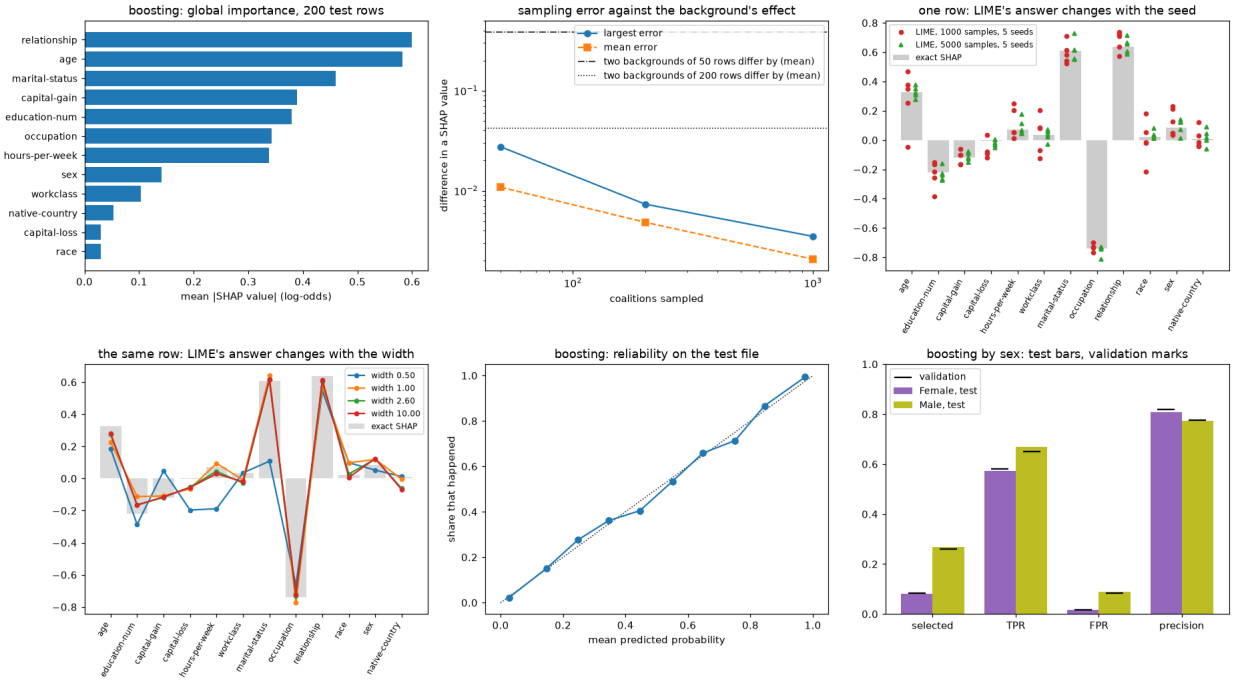
A doctor's annual check-up gives one sheet with blood pressure, cholesterol, weight and notes, each measured the same way every year.

ACTUAL OUTPUT

```
  model             boosting, 200 trees of depth 3, threshold 0.5, trained on 24130 rows
  performance       accuracy 0.8691 [0.8636, 0.8744], AUC 0.9240
  vs baseline       McNemar exact p 4.52e-39: better after Holm
  calibration       ECE 0.0118
  fairness          selected women 0.0804, men 0.2680; TPR women 0.5727, men 0.6688
  tests             4 of 6 significant after Holm: boosting vs logistic (McNemar), sel gap by sex, tpr gap by sex, fpr gap
      by sex
  explanations      largest mean |SHAP|: relationship 0.599, age 0.582, marital-status 0.460
Actual output from this lesson's run.
```

The week's six tools, and what each said today

Day	Tool	On the test file
161	bias, variance, learning curves	training 0.0870 vs test 0.0917 Brier: mild over-fitting
162	hypothesis tests, McNemar	beats logistic regression, p 4.52e-39
163	intervals and the bootstrap	accuracy 0.8691 [0.8636, 0.8744]
164	multiple comparisons	4 of 6 claims survive Holm
165	calibration	ECE 0.0118, about twice its chance floor
166	fairness	women selected 0.0804, men 0.2680; TPR gap +0.0961



Top: mean |SHAP| over 200 test rows; KernelSHAP's error against exact values beside the effect of changing the background; LIME across seeds for one row. Bottom: LIME across widths for the same row; boosting's reliability on the test file; the four rates by sex on test (bars) and validation (marks).

Interview preparation — technical questions

Q: How do you evaluate a model honestly on a test set?

A: Fix the model, threshold and tests on validation data first, score the test set once, and correct for the number of claims.

Q: How do you check a SHAP implementation?

A: Against a model with a known answer: for a linear model the Shapley value is w(x – background mean); here they matched to 2e-16.

Q: What matters most when computing SHAP values?

A: The background: two 50-row backgrounds changed values by up to 0.45 and reordered features; 200 rows brought it to about 0.05.

Q: Is LIME stable?

A: Not at 1,000 samples here: one feature ranged from –0.05 to +0.47 across seeds; more samples and several seeds are needed.

Q: The test ECE is above its chance floor. Do you recalibrate?

A: Not on the test file: a calibrator must be fitted on unseen rows, and the test file is spent; it goes on the card as a known issue.

Interview preparation — theoretical questions

Q: Why do SHAP values add up to the prediction?

A: Efficiency: averaging each feature's marginal contributions over all orders shares out exactly f(x) minus the background mean.

Q: Why does KernelSHAP use a weighted regression?

A: With the Shapley kernel as weights, the regression's solution is the Shapley values; sampling coalitions approximates it.

Q: Does a large SHAP value for sex mean sex causes higher income?

A: No: it describes how the model's output depends on that input, not what would happen to a person.

Q: SHAP or LIME?

A: SHAP when values must add up and be reproducible; LIME as a quick local view, run with several seeds.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> Evaluate like a scientist: fix the plan, look at the test file once, give every number an interval, compare against a baseline with a paired test, correct for the number of claims, and check calibration and groups. Explain with SHAP, checked against a known answer and computed on a large enough background; treat LIME as a sketch.

- Test accuracy 0.8691 [0.8636, 0.8744]; better than the baseline, p 4.52e-39.
- Fairness gaps held on test: selection +0.1876, TPR +0.0961.
- Exact SHAP matched the closed form to 2.2e-16; KernelSHAP within 0.0035.
- Background of 50 rows: values moved by up to 0.4536; 200 rows: 0.0468.

Suggested watch order, if starting from zero

- 1. IML - 04 Shapley - 03 SHAP (SHapley Additive exPlanation) Values (Statistical Learning and Data Science) — SHAP values, as a lecture.
- 2. "Why Should I Trust You?" Explaining the Predictions of Any Classifier (KDD2016 video) — LIME, the talk given with its paper at KDD 2016.

Day 168 preview

- **Week 25 begins: hyperparameter tuning.** Grid search against random search, built from scratch, with a real comparison of how quickly each finds good settings.